Advanced Social, Text and Media Analytics + Lab

Assignment 2

Web Scraping, Text Preprocessing, TF-IDF Representation, and Document Retrieval

The objective of this assignment is to familiarise oneself with web scraping, text preprocessing, TF-IDF-based feature extraction, and document retrieval using cosine similarity. Students will scrape news articles from Inshorts (https://inshorts.com/en/read/), preprocess the text, convert it into TF-IDF vectors, and implement a search system that retrieves the most relevant documents based on a user query.

Prerequisites: Installation

```
!pip install requests beautifulsoup4 nltk scikit-learn

import os

import re

import requests

import nltk

import numpy as np

from bs4 import BeautifulSoup

from nltk.corpus import stopwords

from nltk.stem import WordNetLemmatizer

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.metrics.pairwise import cosine_similarity

nltk.download('stopwords')

nltk.download('wordnet')

nltk.download('omw-1.4')
```

Task 1: ***Data Collection (Web Scraping)***

Scrape around 100 news articles (title + ~60-word description) from Inshorts across different categories (e.g., Sports, Technology, Politics, Entertainment). Each article should be stored in a separate .txt file in ./news_corpus with the naming convention: <category>_<serial_no>.txt

CODE:

```
import requests
from bs4 import BeautifulSoup
import os
import time
categories = {
    "sports": "https://inshorts.com/en/read/sports",
    "technology": "https://inshorts.com/en/read/technology",
    "politics": "https://inshorts.com/en/read/politics",
    "entertainment": "https://inshorts.com/en/read/entertainment"
}
os.makedirs("news_corpus", exist_ok=True)
headers = {
    "User-Agent": "Mozilla/5.0"
}
for category, url in categories.items():
    print("\nProcessing:", category)
    try:
        response = requests.get(
            url,
            headers=headers,
            timeout=15
        )
        print("Status:", response.status_code)
        if response.status_code != 200:
            print("Could not access", category)
            continue
        soup = BeautifulSoup(
```

```
    response.text,
    "html.parser"
)
# Find article headlines
titles = soup.find_all(
    attrs={"itemprop": "headline"}
)
descriptions = soup.find_all(
    attrs={"itemprop": "articleBody"}
)
print("Titles found:", len(titles))
print("Descriptions found:", len(descriptions))
count = 1
for title_tag, desc_tag in zip(
    titles,
    descriptions
):
    title = title_tag.get_text(
        " ",
        strip=True
    )
    description = desc_tag.get_text(
        " ",
        strip=True
    )
    if not title or not description:
        continue
    content = (
```

```
                title +
                "\n" +
                description
            )
            filename = (
                f"{category}_{count}.txt"
            )
            filepath = os.path.join(
                "news_corpus",
                filename
            )
            with open(
                filepath,
                "w",
                encoding="utf-8"
            ) as file:
                file.write(content)
            print("Saved:", filename)
            count += 1
            if count > 25:
                break
        time.sleep(1)
    except Exception as e:
        print(
            "Error:",
            e
        )
print("\nScraping process completed.")
```

OUTPUT:

```
Processing: sports
Status: 200
Titles found: 10
Descriptions found: 10
Saved: sports_1.txt
Saved: sports_2.txt
Saved: sports_3.txt
Saved: sports_4.txt
Saved: sports_5.txt
Saved: sports_6.txt
Saved: sports_7.txt
Saved: sports_8.txt
Saved: sports_9.txt
Saved: sports_10.txt

Processing: technology
Status: 200
Titles found: 10
Descriptions found: 10
Saved: technology_1.txt
Saved: technology_2.txt
Saved: technology_3.txt
Saved: technology_4.txt
Saved: technology_5.txt
Saved: technology_6.txt
Saved: technology_7.txt
Saved: technology_8.txt
Saved: technology_9.txt
Saved: technology_10.txt
```

```
Processing: politics
Status: 200
Titles found: 10
Descriptions found: 10
Saved: politics_1.txt
Saved: politics_2.txt
Saved: politics_3.txt
Saved: politics_4.txt
Saved: politics_5.txt
Saved: politics_6.txt
Saved: politics_7.txt
Saved: politics_8.txt
Saved: politics_9.txt
Saved: politics_10.txt

Processing: entertainment
Status: 200
Titles found: 3
Descriptions found: 3
Saved: entertainment_1.txt
Saved: entertainment_2.txt
Saved: entertainment_3.txt

Scraping process completed.
```

Task 2: Text Preprocessing

Before converting text into numerical features, perform the following preprocessing steps:

Lowercasing: Convert all text to lowercase.

Tokenization: Split sentences into individual words.

Stopword Removal: Remove common stopwords (e.g., "the", "is", "and").

Punctuation & Special Character Removal: Eliminate non-alphabetic symbols.

Stemming/Lemmatization: Reduce words to their base form (e.g., "running" → "run").

CODE:

```
stop_words = set(stopwords.words("english"))

lemmatizer = WordNetLemmatizer()

def preprocess_text(text):

    # Lowercase

    text = text.lower()

    # Keep only alphabets

    text = re.sub(r'[^a-zA-Z\s]', ' ', text)

    # Tokenization
```

```
    tokens = text.split()
    # Stopword removal
    tokens = [
        word for word in tokens
        if word not in stop_words
    ]
    # Lemmatization
    tokens = [
        lemmatizer.lemmatize(word)
        for word in tokens
    ]
    return " ".join(tokens)
```

Read all Document

```
documents = []
filenames = []
corpus_path = "news_corpus"
for filename in sorted(os.listdir(corpus_path)):
    if filename.endswith(".txt"):
        filepath = os.path.join(
            corpus_path,
            filename
        )
        with open(filepath, "r", encoding="utf-8") as file:
            text = file.read()
        documents.append(text)
        filenames.append(filename)
```

Preprocess the documents

```
import textwrap
```

```
processed_documents = [
    preprocess_text(doc)
    for doc in documents
]
for i in range(min(3, len(processed_documents))):
    print("=" * 80)
    print(f"Document {i + 1}")
    print("=" * 80)
    print("\nOriginal:")
    print(textwrap.fill(documents[i], width=80))
    print("\nProcessed:")
    print(textwrap.fill(processed_documents[i], width=80))
    print("\n" + "-" * 80)
```

OUTPUT:

```
================================================================================
Document 1
================================================================================

Original:
No big-budget sequel, leads to needless use of VFX & AI: 'Hanuman Ansh' maker
'Hanuman Ansh' producer Anupriya Nagar revealed that the budget for the sequel
won't increase significantly, saying it might "affect its authenticity". "[It]
wouldn't necessarily make the film better. It could even lead to unnecessary use
of VFX or AI," Anupriya said, adding, "The audience has given us so much love
and support, so we cannot spend money wastefully."

Processed:
big budget sequel lead needless use vfx ai hanuman ansh maker hanuman ansh
producer anupriya nagar revealed budget sequel increase significantly saying
might affect authenticity necessarily make film better could even lead
unnecessary use vfx ai anupriya said adding audience given u much love support
cannot spend money wastefully

--------------------------------------------------------------------------------
```

```
================================================================================
Document 2
================================================================================

Original:
YouTuber Jacob Geller removed from MinnMax podcast YouTuber Jacob Geller has
been removed from podcast MinnMax after past sexual assault findings from his
time at the University of North Carolina resurfaced. Geller confirmed the
findings on Bluesky, saying he was suspended for one semester. He called the
incident "the greatest regret of my life." MinnMax said he won't appear in
future content.

Processed:
youtuber jacob geller removed minnmax podcast youtuber jacob geller removed
podcast minnmax past sexual assault finding time university north carolina
resurfaced geller confirmed finding bluesky saying suspended one semester called
incident greatest regret life minnmax said appear future content

--------------------------------------------------------------------------------
```

Task 3: TF-IDF Vectorization

Convert the preprocessed text into TF-IDF vectors using sklearn.feature_extraction.text.TfidfVectorizer. Each document will be represented as a sparse vector where each dimension corresponds to a unique word in the corpus.

CODE:

```
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(
    processed_documents
)
print("TF-IDF Matrix Shape:")
print(tfidf_matrix.shape)
```

OUTPUT:

```
TF-IDF Matrix Shape:
(33, 871)
```

Task 4: Document Retrieval System

Implement a search system where:

A user inputs a query.

The query is preprocessed and converted into a TF-IDF vector.

Cosine similarity is computed between the query vector and all document vectors.

The top 5 most relevant documents are retrieved and displayed.

CODE:

```
import textwrap
def search_news(query, top_n=5):
    # Preprocess query
    processed_query = preprocess_text(query)
    # Convert query to TF-IDF
    query_vector = vectorizer.transform(
        [processed_query]
```

```
)
# Calculate cosine similarity
similarities = cosine_similarity(
    query_vector,
    tfidf_matrix
).flatten()
# Get top documents
top_indices = similarities.argsort()[::-1][:top_n]
print("\nSearch Query:", query)
print("=" * 70)
for rank, index in enumerate(top_indices, start=1):
    print(f"\nRank {rank}")
    print("File:", filenames[index])
    print(
        "Similarity:",
        round(similarities[index], 4)
    )
    print("\nDocument:")
    # Wrap document text
    wrapped_document = textwrap.fill(
        documents[index][:500],
        width=80
    )
    print(wrapped_document)
    print("-" * 70)
```

Test the search system

Example 1. search_news("artificial intelligence technology")

Output:

```
Search Query: artificial intelligence technology
======================================================================

Rank 1
File: technology_1.txt
Similarity: 0.1557

Document:
AI can never replace judges: CJI Kant Chief Justice of India Surya Kant said
artificial intelligence (AI) may supplement research and support judicial
processes, but can never replace legal acumen or decision-making, which must
always remain with judges. "As our reliance on...AI grows, so too must the
safeguards governing its use," Kant added. He made the remarks while addressing
a plenary session in Uzbekistan.
----------------------------------------------------------------------

Rank 2
File: technology_8.txt
Similarity: 0.0741

Document:
Google sends 4 AI chips into space aboard Musk's SpaceX Google launched a
prototype satellite carrying four Tensor Processing Units (TPUs) into orbit
aboard SpaceX's Falcon 9 rocket. The launch marks the first step in Project
Suncatcher, Google's long-term research effort to explore whether scalable AI
infrastructure, such as data centres, can be hosted in space. During the
mission, the chips will handle simple AI requests using Gemini technology.
----------------------------------------------------------------------
```

```
Rank 3
File: technology_9.txt
Similarity: 0.0

Document:
Elon Musk says 'Instagram is for girls', sparks debate Elon Musk has again
called Instagram a platform "for girls", replying to an X post, which said it
had deleted the Meta-owned app because it was an "inherently female app" with
"zero benefit".  Reacting to it, an X user wrote, "I don't agree. I think
Instagram is a welcoming space for all genders." Another wrote, "What about X?"
----------------------------------------------------------------------

Rank 4
File: technology_6.txt
Similarity: 0.0

Document:
Robotaxis blocking police or firefighters for over 30 mins may be fined in the
US California has passed a law requiring autonomous vehicle companies such as
Tesla, Waymo and Zoox to provide local support when robotaxis cause problems or
block emergency responders. Companies could face penalties if a robotaxi
obstructs police or firefighters for over 30 minutes. The law aims to improve
safety and response times after several incidents involving disabled or
disruptive robotaxis.
----------------------------------------------------------------------
```

```
Rank 5
File: technology_7.txt
Similarity: 0.0

Document:
ChatGPT now lets users virtually try on clothes, accessories OpenAI has
introduced a new shopping feature in ChatGPT, allowing users to virtually try on
clothes and accessories. Users can upload a selfie or full-body photo and select
the new option on eligible shopping results. They can also upload a photo of an
item or a shopping screenshot and ask ChatGPT to show how it looks on them.
----------------------------------------------------------------------
```

Example 2. search_news("cricket match India")

Output:

```
Search Query: cricket match India
======================================================================

Rank 1
File: sports_6.txt
Similarity: 0.1069

Document:
Chopra backs Nabi for 3rd ODI despite poor debut Aakash Chopra backed Auqib Nabi
to play the 3rd ODI vs West Indies in Mullanpur, despite conceding 84 runs on
debut in Guwahati. Chopra noted Nabi's style suits red-ball cricket more but
urged India to persist with him since he was picked for his batting ability at
number eight.
----------------------------------------------------------------------

Rank 2
File: sports_9.txt
Similarity: 0.0912

Document:
Parthiv Patel takes dig at Pakistan over 2027 World Cup Former India
wicketkeeper-batter Parthiv Patel took a dig at Pakistan while discussing the
newly announced format and schedule of the 2027 ICC ODI World Cup. Patel joked
that Pakistan would be pleased with the format and confident of making it to the
second round of the tournament. The International Cricket Council (ICC)
announced the schedule for the 2027 ODI World Cup.
----------------------------------------------------------------------
```

```
Rank 3
File: sports_8.txt
Similarity: 0.0798

Document:
Asian Games: Historic double final draw praise from hockey society Indian hockey
has achieved a historic milestone at the Asian Games 2026, with both the Indian
men's and women's teams reaching the finals after winning their respective semi-
final matches. "For the first time since 1998, both the Indian men's and women's
hockey teams have reached the final at the same edition of the Asian Games,"
said Iqbal Singh Sandhu, CEO.
----------------------------------------------------------------------

Rank 4
File: sports_4.txt
Similarity: 0.0589

Document:
Aakash Chopra sees Ishan Kishan as future T20I vice-captain Former India batter
Aakash Chopra predicted Ishan Kishan could soon become India's T20I vice-
captain. His remarks followed Kishan's unbeaten 80 off 46 balls against Sri
Lanka in the Asian Games 2026 semi-final. Kishan leads T20I run-scoring charts
in 2026 with 985 runs in 28 innings at a strike rate of 171.30.
----------------------------------------------------------------------

Rank 5
File: sports_3.txt
Similarity: 0.0575

Document:
Kaif: Reddy can't replace Hardik; back Jadeja Former cricketer Mohammad Kaif
said Nitish Kumar Reddy can never truly replace Hardik Pandya, calling Reddy
primarily a batter. He questioned India's decision to drop Reddy without a
batting chance in the 2nd ODI vs West Indies. Kaif also backed Jadeja and urged
India to strengthen bowling depth ahead of the 2027 World Cup.
----------------------------------------------------------------------
```